TRIVIAL DUALITY FOR SYSTEMS OF PERFECT OBJECTS: RHOM(HOCOLIM K_N, E) = RLIM E TENSOR K_N^DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- For a perfect complex K_n with dual K_n^dual = RHom(K_n, A) one has E tensor K_n^dual = RHom_A(K_n, E) for every E. invoked at proof lines 19251-19253. Turns the right-hand side into a limit of Hom complexes.
- The derived colimit of a system is the cone of the canonical map from the direct sum to itself. invoked at statement line 19242. Identifies K with the cone of a map of direct sums, matching the Rlim triangle.
- The internal-hom adjunction formula Hom(K, RHom(L, M)) = Hom(K tensor L, M). invoked at proof lines 19263-19264. Reduces Hom out of a direct sum to a product of Hom complexes.

Source reference. Original definition 090Z: sources/stacks-chow/tags.txt, source lines 11427–11427. Cited at statement line 19242 to define the derived colimit K = hocolim K_n; the definition is used inside the counted span.

Source reference. Original lemma 07VI: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19176–19176. Cited at proof line 19251 to rewrite E tensor K_n^dual as RHom_A(K_n, E); the conclusion is used inside the counted span.

Source reference. Original equation 0A63: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18654–18654. Cited at proof line 19263 to reduce the Hom out of the direct sum to a product of Hom complexes; the formula is used inside the counted span.

Lemma 0.2. *Let A be a ring. Let $(K_n)_{n\in\mathbf{N}}$ be a system of perfect objects of $D(A)$. Let $K = \text{hocolim} K_n$ be the derived colimit (Derived Categories, Definition 090Z). Then for any object E of $D(A)$ we have*

$$R\operatorname{Hom}_A(K, E) = R\lim E \otimes^{\mathbf{L}}_A K_n^\vee$$

where $(K_n^\vee)$ is the inverse system of dual perfect complexes.

Proof. By Lemma 07VI we have $R\lim E \otimes^{\mathbf{L}}_A K_n^\vee = R\lim R\operatorname{Hom}_A(K_n, E)$ which fits into the distinguished triangle

$$R\lim R\operatorname{Hom}_A(K_n, E) \to \prod R\operatorname{Hom}_A(K_n, E) \to \prod R\operatorname{Hom}_A(K_n, E)$$

Because K similarly fits into the distinguished triangle $\bigoplus K_n \to \bigoplus K_n \to K$ it suffices to show that $\prod R\operatorname{Hom}_A(K_n, E) = R\operatorname{Hom}_A(\bigoplus K_n, E)$. This is a formal

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

consequence of (0A63) and the fact that derived tensor product commutes with direct sums. □